Software Requirements Specification

for

SMART HOSPITAL MANAGEMENT SYSTEM

Version 1.0 approved

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for Version 1.0 of the Smart Hospital Management System (SHMS). SHMS is a web-based application intended to digitize and streamline core hospital operations, including patient registration, electronic health records, appointment scheduling, billing and insurance processing, and in-patient bed/ward allocation. This document covers the complete scope of the system as a single, integrated product; it does not describe any external systems with which SHMS may later be integrated.

1.2 Product Scope

The Smart Hospital Management System is a centralized, role-based software platform designed to replace manual, paper-based hospital workflows with a secure digital system. The system's key objectives are to: reduce patient waiting time and administrative overhead; maintain accurate, accessible, and secure electronic health records; enable efficient appointment scheduling between patients and doctors; automate billing, insurance claim processing, and online payments; and optimize the allocation of beds and wards for in-patient care. By consolidating these functions into one platform accessible to all hospital stakeholders through role-appropriate interfaces, SHMS aims to improve operational efficiency, reduce errors, and enhance the overall quality of patient care.

1.3 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- Wiegers, Karl E., Software Requirements Specification (SRS) Template, 1999.
- Health Insurance Portability and Accountability Act (HIPAA) — reference standard for healthcare data privacy (region-specific equivalent to be confirmed; see Appendix C).
- <Institution's Software Engineering Course Case Study Brief — Smart Hospital Management System>.

2. Overall Description

2.1 Product Perspective

SHMS is a new, self-contained, web-based product intended to replace existing manual or partially digitized hospital record-keeping processes. It is not a component of a larger pre-existing system, although it is designed to allow future integration with external systems such as insurance provider APIs, payment gateways, pharmacy inventory systems, and laboratory equipment interfaces. The system follows a three-tier architecture consisting of a presentation layer (web/mobile client), an application/business logic layer (server-side services), and a data layer (centralized database storing patient, appointment, billing, and administrative records).

The following users interact with the system, each through a dedicated, role-specific interface, all

connecting to a single shared backend and database:

- Patients — register, book appointments, and view records/bills remotely.
- Clinical staff (Doctors, Nurses, Lab Technicians) — access and update patient care data.
- Administrative/support staff (Receptionists, Pharmacists) — manage day-to-day front-office and pharmacy operations.
- Administrators — configure and oversee the entire system.

2.2 Product Functions

The major functions provided by SHMS are summarized below; detailed requirements for each are provided in Section 4.

- Role-Based Access Control (RBAC) — secure login and role-specific dashboards for Patient, Doctor, Nurse, Receptionist, Pharmacist, Lab Technician, and Administrator.
- Patient Registration and Electronic Health Records (EHR) — capture patient demographics and maintain a longitudinal digital medical history.
- Doctor Appointment Booking and Scheduling — allow patients to book, reschedule, or cancel appointments, and allow doctors to manage their availability.
- Billing, Insurance, and Online Payment Management — generate invoices, process insurance claims, and accept online payments.
- Bed and Ward Allocation (In-Patient Management) — track bed availability and manage admission, transfer, and discharge of in-patient patients.

2.3 User Classes and Characteristics

The table below identifies the user classes anticipated for SHMS. All user classes are considered important; Patients, Doctors, and Receptionists are the most frequent users and are prioritized for usability, while Administrator functions are prioritized for security and control.

User Class	Characteristics	Primary Functions Used
Patient	General public; varying levels of computer literacy; accesses system remotely via web/mobile.	Registration, view EHR, book/cancel appointments, view bills, make online payments.
Doctor	Medical professional; moderate to high technical proficiency; time-constrained.	View patient EHR, manage schedule/appointments, record diagnoses and prescriptions.
Nurse	Clinical staff supporting doctors and in-patient care; ward-level access.	Update patient vitals, manage bed/ward status, assist with EHR entries.
Receptionist	Front-desk administrative staff; frequent daily use.	Patient registration, appointment scheduling, bed allocation coordination.
Pharmacist	Dispenses medication; needs access to prescriptions.	View prescriptions, manage inventory, dispense medication,

User Class	Characteristics	Primary Functions Used
		update stock.
Lab Technician	Performs diagnostic tests; updates results into EHR.	Receive test orders, upload lab results, notify doctors of completion.
Administrator	IT/hospital management staff; highest privilege level.	User/role management, system configuration, reporting, auditing, billing oversight.

2.4 Operating Environment

SHMS shall operate as a responsive web application accessible via modern web browsers (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari — current and previous major versions) on desktop and mobile devices. The application shall be hosted on a cloud or on-premises application server running a supported operating system (e.g., Linux-based server OS) and shall connect to a relational database management system. Patient-facing features should also be optimized for mobile browser access; a dedicated mobile application is outside the scope of Version 1.0 but may be considered for future releases.

2.5 Design and Implementation Constraints

- The system must comply with applicable healthcare data privacy and protection regulations governing the storage and transmission of patient health information.
- All patient health data must be encrypted at rest and in transit.
- Online payment processing must be delegated to a PCI-DSS compliant third-party payment gateway; SHMS shall not store raw card details.
- The system shall be developed using a relational database to preserve referential integrity between patients, appointments, billing, and bed-allocation records.
- The user interface must support role-based views such that users can only see menu options and data relevant to their assigned role.
- The system must be available 24/7 with minimal scheduled downtime, given the critical nature of hospital operations.

2.6 User Documentation

The following user documentation will be delivered with the software: a Patient User Guide (registration, appointment booking, bill payment); a Clinical Staff User Guide (EHR access and updates for Doctors, Nurses, and Lab Technicians); a Front-Office User Guide (registration and bed allocation for Receptionists); a Pharmacist User Guide; and an Administrator Manual covering user management, configuration, and reporting. Context-sensitive online help and tooltips will be provided within each role's dashboard.

2.7 Assumptions and Dependencies

- It is assumed that all hospital staff using the system will have basic computer literacy and access to a device with a modern web browser and internet connectivity.

- It is assumed that a stable internet connection is available at all hospital workstations and for remote patient access.
- The system depends on a third-party payment gateway service for processing online payments; changes to that gateway's API may require corresponding changes to SHMS.
- The system depends on insurance providers/TPAs supporting a compatible mechanism (API or file-based) for claim submission and status verification; the exact list of supported providers is TBD (see Appendix C).
- It is assumed that the hospital will provide accurate, up-to-date master data for wards, beds, departments, and staff roles prior to system go-live.

3. External Interface Requirements

3.1 User Interfaces

SHMS shall provide seven distinct, role-based dashboards (Patient, Doctor, Nurse, Receptionist, Pharmacist, Lab Technician, Administrator), each exposing only the menu items and data relevant to that role. All interfaces shall share a consistent visual style guide (color palette, typography, iconography) and a common header containing the hospital logo, logged-in user's name/role, a notifications icon, and a logout control. Every screen shall provide a visible "Help" option and standardized, human-readable error messages (e.g., "This time slot is no longer available. Please select another slot.") rather than raw system error codes. Forms shall perform client-side validation with inline error indicators before submission. Detailed wireframes and screen layouts will be documented in a separate UI Specification.

3.2 Hardware Interfaces

SHMS is a software-only product with no dedicated hardware of its own. It shall run on standard desktop computers, laptops, tablets, and smartphones equipped with a supported web browser. Where available, the Receptionist and Pharmacist interfaces should support integration with standard peripherals such as barcode/RFID scanners (for patient ID cards or bed/asset tags) and receipt/invoice printers, communicating via standard USB or network drivers provided by the operating system; exact hardware models are TBD (see Appendix C).

3.3 Software Interfaces

- Database Management System: a relational DBMS (e.g., PostgreSQL or MySQL) shall store all patient, appointment, billing, and bed-allocation data.
- Payment Gateway: an external, PCI-DSS compliant payment gateway API (provider TBD, see Appendix C) shall be used to process online card/UPI payments and return transaction status to SHMS.
- Insurance/TPA Interface: SHMS shall exchange claim submission and status data with external insurance provider or TPA systems via API or a secure file-based interface (exact protocol TBD).
- Notification Services: SHMS shall integrate with an SMS gateway and an email service (e.g., SMTP) to send appointment reminders, billing notifications, and account alerts.
- Operating System: the server-side application shall be compatible with a modern Linux-based server operating system; client interfaces shall be OS-independent, accessed via a web browser.

3.4 Communications Interfaces

All communication between the client (browser) and the SHMS application server shall use HTTPS (TLS 1.2 or higher) to ensure data confidentiality and integrity. Server-to-server communication with the payment gateway and insurance/TPA systems shall similarly use encrypted HTTPS/REST API calls, authenticated via API keys or OAuth 2.0 tokens as required by each third party. Email notifications shall be sent via SMTP over TLS; SMS notifications shall be sent via the SMS gateway provider's HTTPS API. Data formatting for all API exchanges shall use JSON unless a specific external system requires an alternate format (e.g., XML or HL7 for future clinical interoperability).

4. System Features

This section organizes the functional requirements of SHMS by system feature, corresponding to the five major capability areas defined for this case study.

4.1 Role-Based Access Control (RBAC)

4.1.1 Description and Priority

Provides secure authentication and authorization so that each of the seven user types (Patient, Doctor, Nurse, Receptionist, Pharmacist, Lab Technician, Administrator) can access only the functions and data appropriate to their role. Priority: High.

Priority	Benefit	Penalty	Cost	Risk
High	9	9	5	6

4.1.2 Stimulus/Response Sequences

- User enters credentials and submits the login form → system validates credentials and redirects to the role-appropriate dashboard, or displays an error if invalid.
- Administrator creates or edits a staff account and assigns a role → system stores the account and grants the corresponding role-based permissions on next login.
- User attempts to access a URL/function outside their role's permission set → system denies access and displays an "Access Denied" message.

4.1.3 Functional Requirements

REQ-4.1.1: The system shall require all users to authenticate with a unique username/ID and password (or hospital-issued patient ID for Patients) before accessing any functionality.

REQ-4.1.2: The system shall support exactly seven user roles: Patient, Doctor, Nurse, Receptionist, Pharmacist, Lab Technician, and Administrator, each with a distinct, predefined set of permissions.

REQ-4.1.3: The system shall restrict each user to functions and data authorized for their assigned role and shall deny and log any unauthorized access attempt.

REQ-4.1.4: Only Administrators shall be able to create, modify, deactivate, or delete staff user accounts and assign/change their roles.

REQ-4.1.5: The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts and shall notify the user via registered email or SMS.

REQ-4.1.6: The system shall allow users to reset a forgotten password via a secure, time-limited link sent to their registered email or mobile number.

REQ-4.1.7: The system shall automatically log out an inactive session after 15 minutes of inactivity.

4.2 Patient Registration and Electronic Health Records (EHR)

4.2.1 Description and Priority

Allows new patients to register (in person via Receptionist or self-service online) and maintains a comprehensive digital medical history for each patient, accessible to authorized clinical staff. Priority: High.

Priority	Benefit	Penalty	Cost	Risk
High	9	8	6	5

4.2.2 Stimulus/Response Sequences

- Receptionist or Patient submits a new registration form → system validates the data, creates a unique patient ID, and generates an initial (empty) EHR.
- Doctor completes a consultation and enters diagnosis, prescription, and notes → system appends the entry to the patient's EHR with a timestamp and the doctor's ID.
- Lab Technician uploads a test result → system attaches the result to the patient's EHR and notifies the ordering doctor.
- Authorized clinical user searches for a patient by ID or name → system displays matching patient records and their EHR summary.

4.2.3 Functional Requirements

REQ-4.2.1: The system shall allow a Receptionist or a self-registering Patient to create a new patient profile capturing name, date of birth, gender, contact details, address, emergency contact, and government-issued ID (where applicable).

REQ-4.2.2: The system shall automatically generate a unique Patient ID upon successful registration and shall prevent duplicate registration using matching name and date of birth combined with contact details, flagging potential duplicates for Receptionist review.

REQ-4.2.3: The system shall maintain, for each patient, a chronological EHR containing consultation notes, diagnoses, prescriptions, lab results, allergies, and past admission history.

REQ-4.2.4: The system shall allow Doctors and Nurses to view a patient's complete EHR, and shall allow only Doctors to add or amend diagnosis and prescription entries.

REQ-4.2.5: The system shall allow Lab Technicians to upload diagnostic test results directly into the relevant patient's EHR and to notify the requesting Doctor upon completion.

REQ-4.2.6: The system shall allow Patients to view (read-only) their own EHR summary, including diagnoses, prescriptions, and lab results, through their patient portal.

REQ-4.2.7: The system shall maintain an immutable audit trail of every view and modification made to a patient's EHR, recording the user, timestamp, and action taken.

REQ-4.2.8: The system shall retain EHR data for a minimum retention period as required by applicable

healthcare recordkeeping regulations (exact period TBD, see Appendix C).

4.3 Doctor Appointment Booking and Scheduling

4.3.1 Description and Priority

Enables patients to view doctor availability and book, reschedule, or cancel appointments, while allowing doctors to define and manage their own availability and daily schedule. Priority: High.

Priority	Benefit	Penalty	Cost	Risk
High	8	7	5	4

4.3.2 Stimulus/Response Sequences

- Patient selects a department/doctor and an available time slot → system reserves the slot and generates a confirmation with appointment details.
- Patient requests to cancel or reschedule an existing appointment → system releases or updates the slot and notifies the doctor and patient.
- Doctor updates their weekly availability or marks leave → system blocks the corresponding slots from patient booking.
- Receptionist books a walk-in appointment on behalf of a patient → system reserves the next available slot for the requested doctor.

4.3.3 Functional Requirements

REQ-4.3.1: The system shall allow Patients and Receptionists to search for doctors by department, specialization, or name and view their available appointment slots.

REQ-4.3.2: The system shall allow a Patient to book an available appointment slot, which shall immediately become unavailable to other patients (no double-booking).

REQ-4.3.3: The system shall send an appointment confirmation to the Patient via email and/or SMS immediately upon successful booking.

REQ-4.3.4: The system shall allow a Patient to reschedule or cancel an appointment up to a configurable minimum notice period (default: 2 hours) before the scheduled time.

REQ-4.3.5: The system shall allow Doctors to define, and subsequently modify, their recurring weekly availability and to mark specific dates/times as unavailable (e.g., leave, emergency).

REQ-4.3.6: The system shall send an automated reminder notification to the Patient (e.g., 24 hours and 2 hours before) for each upcoming appointment.

REQ-4.3.7: The system shall allow Doctors and Receptionists to view a daily/weekly schedule of all booked appointments for a given doctor.

REQ-4.3.8: The system shall allow a Doctor or Receptionist to mark an appointment as Completed, No-Show, or Cancelled after the scheduled time has passed.

4.4 Billing, Insurance, and Online Payment Management

4.4.1 Description and Priority

Generates itemized bills for consultations, tests, procedures, and admissions; supports submission and tracking of insurance claims; and enables patients to pay outstanding balances online. Priority: High.

Priority	Benefit	Penalty	Cost	Risk
High	8	8	6	6

4.4.2 Stimulus/Response Sequences

- A billable service (consultation, test, admission) is completed → system automatically adds the corresponding charge to the patient's active bill.
- Patient chooses to pay a bill online → system redirects to the payment gateway and updates the bill status upon confirmation.
- Receptionist or Patient submits insurance details for a bill → system generates a claim record and forwards it to the relevant insurer/TPA for approval.
- Insurer/TPA returns a claim decision → system updates the claim status and adjusts the patient's payable balance accordingly.

4.4.3 Functional Requirements

REQ-4.4.1: The system shall automatically generate an itemized bill for each patient encounter, including consultation fees, tests, procedures, medication, and (for in-patients) ward/bed charges.

REQ-4.4.2: The system shall allow Patients to view their current and historical bills, including a breakdown of charges, through the patient portal.

REQ-4.4.3: The system shall allow a Patient to pay an outstanding bill online via an integrated, PCI-DSS compliant payment gateway supporting at least credit/debit card and UPI/net-banking payment methods.

REQ-4.4.4: The system shall update the bill status to Paid and issue a digital receipt immediately upon confirmation of a successful online payment.

REQ-4.4.5: The system shall allow Patients or Receptionists to submit insurance policy details against a bill and generate an insurance claim request.

REQ-4.4.6: The system shall track the status of each insurance claim (e.g., Submitted, Under Review, Approved, Partially Approved, Rejected) and notify the Patient of status changes.

REQ-4.4.7: The system shall recalculate the patient's payable balance automatically based on the approved insurance coverage amount once a claim decision is received.

REQ-4.4.8: The system shall allow Administrators to generate financial reports summarizing billing, payments received, and outstanding dues over a specified date range.

REQ-4.4.9: The system shall never store complete, unmasked payment card numbers; all card data shall be handled exclusively by the certified payment gateway.

4.5 Bed and Ward Allocation (In-Patient Management)

4.5.1 Description and Priority

Manages real-time availability of beds across hospital wards and supports the admission, transfer, and discharge workflow for in-patients. Priority: Medium.

Priority	Benefit	Penalty	Cost	Risk
Medium	7	6	5	5

4.5.2 Stimulus/Response Sequences

- Doctor recommends admission for a patient → system displays available beds matching the required ward type for Receptionist/Nurse selection.
- Receptionist or Nurse allocates a specific bed to the patient → system marks the bed as Occupied and links it to the patient's active admission record.
- Doctor or Nurse requests a ward transfer for a patient → system releases the current bed and reserves a new bed in the target ward, if available.
- Doctor approves discharge → system marks the bed as Available (pending housekeeping/cleaning status) and finalizes the in-patient billing.

4.5.3 Functional Requirements

REQ-4.5.1: The system shall maintain a real-time inventory of all beds, organized by ward (e.g., General, ICU, Maternity, Pediatric), with status of each bed as Available, Occupied, or Under Maintenance.

REQ-4.5.2: The system shall allow a Doctor to initiate an admission request for a patient, specifying the required ward type.

REQ-4.5.3: The system shall allow a Receptionist or Nurse to view available beds matching a requested ward type and to allocate a specific bed to a patient, creating an active admission record.

REQ-4.5.4: The system shall prevent a bed marked as Occupied from being allocated to more than one patient at a time.

REQ-4.5.5: The system shall allow Nurses and Doctors to record and update a patient's ward transfer, automatically releasing the previous bed and reserving the new one.

REQ-4.5.6: The system shall allow a Doctor to approve a patient's discharge, upon which the system shall mark the associated bed as Available and trigger generation of the final in-patient bill.

REQ-4.5.7: The system shall allow Administrators and Receptionists to view a hospital-wide, real-time dashboard of bed occupancy and availability by ward.

REQ-4.5.8: The system shall alert Receptionists and ward-in-charge Nurses when a ward's bed occupancy exceeds a configurable capacity threshold (default: 90%).

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall support at least 500 concurrent users without degradation of response time

below acceptable thresholds (exact peak load to be confirmed; see Appendix C).

- 95% of page navigation and search requests shall complete within 3 seconds under normal load conditions.
- Appointment slot booking confirmation shall be returned to the user within 5 seconds of submission.
- The bed/ward availability dashboard shall reflect allocation or discharge updates within 2 seconds (near real-time) of the underlying change.

5.2 Safety Requirements

- The system shall never permanently delete patient EHR data; records may only be archived or marked inactive, preserving an audit trail.
- The system shall provide automated daily backups of all patient, billing, and bed-allocation data, stored in a geographically separate location from the primary database.
- In the event of a system outage, critical bed-availability and patient-identification data shall be recoverable within a Recovery Time Objective (RTO) of 4 hours.
- The system shall display clear alerts for known patient allergies whenever a Doctor is entering a new prescription for that patient.

5.3 Security Requirements

- All patient health data and payment information shall be encrypted both at rest (e.g., AES-256) and in transit (TLS 1.2 or higher).
- The system shall enforce role-based access control as defined in Section 4.1 for every data access and transaction.
- The system shall maintain a tamper-evident audit log of all access to and modification of patient EHR, billing, and admission records, retained for a minimum of one year.
- Passwords shall be stored using a salted, industry-standard one-way hashing algorithm (e.g., bcrypt); plaintext passwords shall never be stored or logged.
- The system shall comply with applicable healthcare data privacy regulations governing the collection, storage, and disclosure of patient health information (specific regulation TBD, see Appendix C).

5.4 Software Quality Attributes

- Usability: role-specific dashboards shall be learnable by a new staff member with less than 2 hours of training; patient-facing screens shall be usable without any training.
- Availability: the system shall maintain 99.5% uptime, excluding scheduled maintenance windows communicated at least 48 hours in advance.
- Reliability: the system shall correctly process at least 99.9% of appointment bookings and billing transactions without data loss or corruption.
- Maintainability: the system shall be built using a modular architecture that allows individual features (e.g., billing, EHR) to be updated independently with minimal impact on other modules.
- Scalability: the system architecture shall support horizontal scaling to accommodate additional hospital branches or increased user load without a redesign.

- Interoperability: the system should be designed to support future integration with external clinical systems using standard healthcare data exchange formats (e.g., HL7/FHIR).

5.5 Business Rules

- Only Administrators may create, modify, or deactivate staff user accounts and assign roles.
- Only Doctors may enter or modify diagnosis and prescription entries in a patient's EHR.
- A bed may be allocated to only one patient at any given time; concurrent allocation requests shall be resolved on a first-confirmed basis.
- An appointment slot, once booked, shall not be booked by another patient unless it is first cancelled or has passed its scheduled time.
- A patient's final bill for an in-patient stay shall not be closed/finalized until the attending Doctor has formally approved discharge.
- Insurance claims shall only be submitted for bills that have been finalized (i.e., no further billable charges pending).

6. Other Requirements

Internationalization: the system should support display of the user interface in English by default, with the underlying architecture designed to allow additional language packs to be added in future releases without significant redesign.

Legal/Regulatory: the system's data retention, consent, and disclosure practices for patient health information shall comply with applicable national and regional healthcare data protection laws; the specific applicable regulation is to be confirmed with hospital legal counsel (see Appendix C).

Database Requirements: the production database shall be backed up nightly with point-in-time recovery capability, and shall enforce referential integrity across patient, appointment, billing, and bed-allocation tables as described in Section 2.5.

Appendix A: Glossary

Term	Definition
EHR	Electronic Health Record — a digital record of a patient's medical history, diagnoses, medications, treatment plans, and lab results.
RBAC	Role-Based Access Control — a mechanism restricting system access based on the role assigned to a user.
OP	Out-Patient — a patient who receives treatment without an overnight hospital stay.
IP	In-Patient — a patient admitted to the hospital and allocated a bed/ward for treatment lasting one or more days.
Ward	A defined section of the hospital containing beds for a specific department or category of patients (e.g., General, ICU, Maternity).

Term	Definition
Payment Gateway	A third-party service that authorizes and processes online credit/debit card or UPI payments.
TPA	Third-Party Administrator — an entity that processes insurance claims on behalf of an insurance company.
SHMS	Smart Hospital Management System — the software product specified in this document.
Admin	Administrator — the user role responsible for system configuration, user management, and oversight.

Appendix B: Analysis Models

The following analysis models support this SRS and will be maintained as separate design artifacts:

- Use Case Diagram — showing interactions between all seven actor roles and the major system functions described in Section 4.
- Entity-Relationship (ER) Diagram — modeling core entities including Patient, EHR Record, Appointment, Doctor, Bill, Insurance Claim, Ward, and Bed, and their relationships.
- Data Flow Diagram (Level 0 and Level 1) — illustrating the flow of patient, appointment, billing, and bed-allocation data between system processes and external entities (patients, insurers, payment gateway).
- State-Transition Diagram — modeling the lifecycle of an in-patient bed (Available → Occupied → Under Maintenance → Available) and of an appointment (Booked → Completed / Cancelled / No-Show).

Appendix C: To Be Determined List

The following items remain open and shall be tracked to closure before or during development:

#	TBD Item	Section
1	Final selection of the payment gateway provider (e.g., Razorpay, Stripe, PayU) for online payment processing.	3.3 / 4.4
2	Exact list of insurance providers and TPAs to be integrated in the first release.	4.4
3	Specific regulatory/compliance standard to be followed for health data (e.g., HIPAA, local health data protection law).	5.3
4	Maximum concurrent user load for peak-hour performance testing.	5.1
5	Final hardware specification for barcode/RFID-based bed and asset tracking, if adopted.	3.2